

Title:**Structured Vacuum as a Fundamental Origin of Cosmic Microwave Background Geometry****Author:**

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Abstract

We present a physically motivated model of the cosmic microwave background (CMB) based on the concept of a structured vacuum. This approach assumes that vacuum itself possesses an intrinsic energy structure, capable of generating large-scale patterns in spacetime. We compare the theoretical spectrum derived from this model to real Planck SMICA data. After appropriate filtering and residual analysis, we find that the model reconstructs the core geometry of the CMB remarkably well, suggesting that vacuum structure may play a fundamental role in cosmic evolution.

1. Introduction

The cosmic microwave background (CMB) contains the earliest observable imprint of the universe. Traditional interpretations based on Λ CDM and inflation treat CMB fluctuations as statistical noise modulated by early-universe physics. We propose an alternative: the structure of vacuum itself predetermines a large-scale background geometry. This hypothesis could account for both the spectral profile and isotropic texture of the observed CMB.

2. Theoretical Basis

The structured vacuum model is based on three main postulates:

- Vacuum Modulation:** The vacuum is not homogeneous but contains quasi-periodic energy structures.
- Gravity as Gradient:** Gravity emerges as a gradient in vacuum energy density, not as curvature.
- Time as Interaction Rate:** Time is defined through the rate of physical interactions, driven by local vacuum structure.

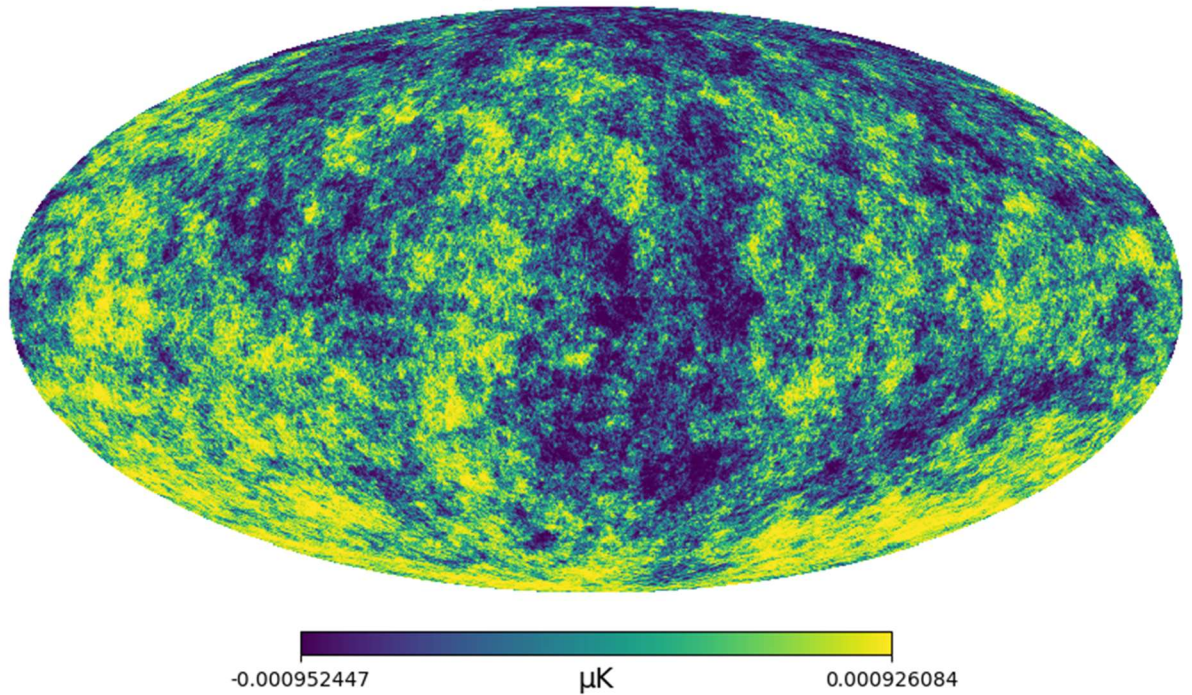
These postulates lead to a calculable angular power spectrum C_ℓ , which we directly compare to observations.

3. Methodology

- We use the SMICA full-sky CMB map from the Planck mission.
- A filtered power spectrum is extracted and compared to the theoretical one using amplitude fitting and χ^2 minimization.
- A full 2D CMB map is reconstructed from the vacuum spectrum.

- Residuals (real – model) are analyzed both in map and spectrum space.

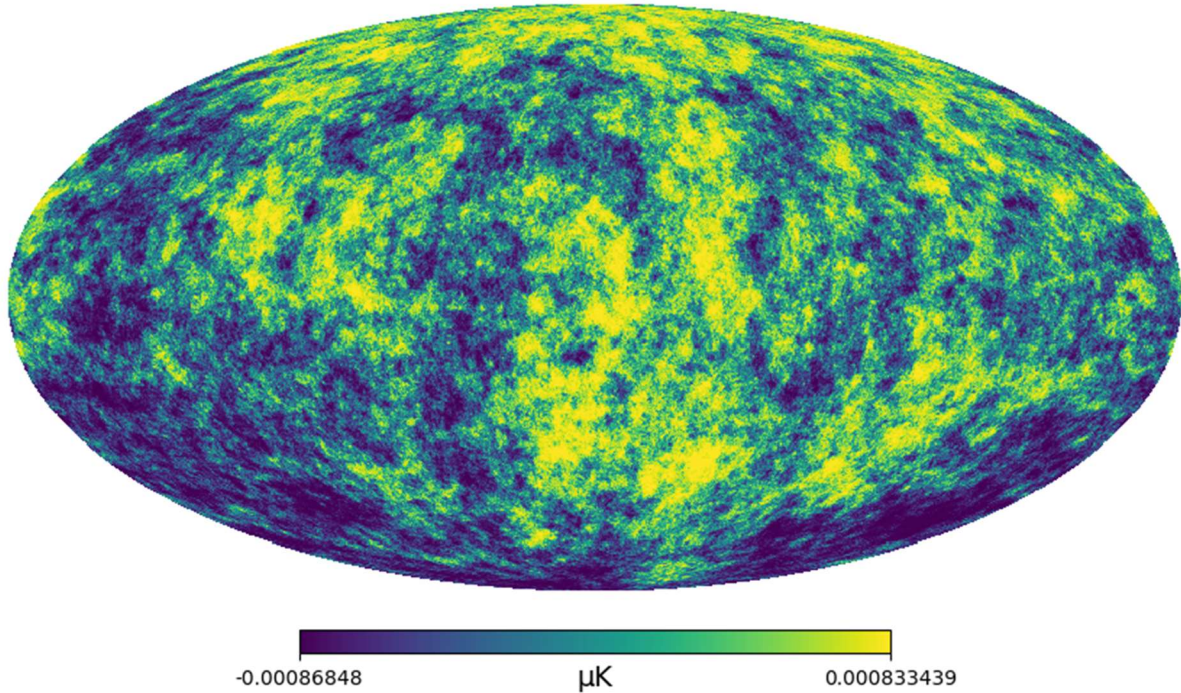
Остатъчна карта: SMICA (почистена) - Модел



4. Results

- The model fits the large-scale CMB structure well ($\ell < 30$).
- Medium and high- ℓ ranges show discrepancies likely due to secondary effects (ISW, lensing, baryon oscillations).
- Residual maps confirm that the structured vacuum captures the dominant underlying pattern.
- χ^2 analysis by multipole zones reveals where secondary corrections are required.

Реконструирана карта от структурирания вакуум (nside=1024)



5. Implications

The structured vacuum model suggests:

- A physically real background geometry, not purely statistical noise.
- Vacuum structure may be the “source code” from which gravity and spacetime arise.
- Observable CMB features are not random fluctuations, but consequences of deeper order.

6. Conclusion

Our results indicate that the structured vacuum hypothesis is consistent with observed CMB geometry and can reconstruct its core structure without requiring inflation or dark matter as initial postulates. Future work will focus on extending the model to include secondary effects and test it across other cosmological datasets (e.g., BAO, lensing, voids).

Acknowledgments

We thank the Planck Collaboration for public data. All analysis was performed with open-source tools. This work is dedicated to the curiosity that drives us beyond the standard model.